1. If a prime number p divides ab then either p divides a or p divides b, that is p divides at least one of them.
2. If a composite number n divides ab, then n neither divide a nor b.
   For example, 6 divides 4×3 but 6 neither divide 4 nor 3.

Example 2.7 In the given factorisation, find the numbers m and n.

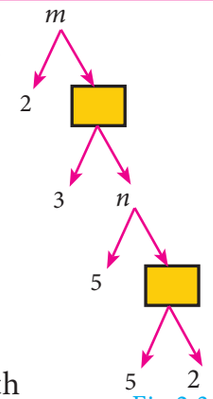


Fig.2.3

Solution Value of the first box from bottom $= 5 \times 2 = 10$

Value of n $= 5 \times 10 = 50$

Value of the second box from bottom $= 3 \times 50 = 150$

Value of m $= 2 \times 150 = 300$

Thus, the required numbers are $m = 300,\ n = 50$

Example 2.8 Can the number 6^n, n being a natural number end with the digit 5? Give reason for your answer.

Solution Since $6^n = (2 \times 3)^n = 2^n \times 3^n$,

2 is a factor of 6^n. So, 6^n is always even.

But any number whose last digit is 5 is always odd.

Hence, 6^n cannot end with the digit 5.

Progress Check

1. Let m divides n. Then GCD and LCM of m, n are ____ and ____.
2. The HCF of numbers of the form 2^m and 3^n is _____.

Example 2.9 Is $7 \times 5 \times 3 \times 2 + 3$ a composite number? Justify your answer.

Solution Yes, the given number is a composite number, because

$$7 \times 5 \times 3 \times 2 + 3 = 3 \times \left(7 \times 5 \times 2 + 1\right) = 3 \times 71$$

Since the given number can be factorized in terms of two primes, it is a composite number.

Example 2.10 'a' and 'b' are two positive integers such that $a^b \times b^a = 800$. Find 'a' and 'b'.

Solution The number 800 can be factorized as

$$800 = 2 \times 2 \times 2 \times 2 \times 2 \times 5 \times 5 = 2^5 \times 5^2$$

Hence, $a^b \times b^a = 2^5 \times 5^2$

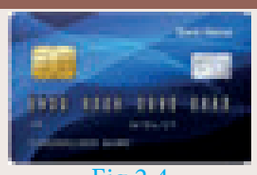
This implies that $a = 2$ and $b = 5$ (or) $a = 5$ and $b = 2$.

Thinking Corner

Can you think of positive integers a, b such that $a^b = b^a$?

Activity 3

Can you find the 4-digit pin number '*pqrs*' of an ATM card such that $p^2 \times q^1 \times r^4 \times s^3 = 3,15,000$?


Fig.2.4

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